AdifySkills

n8n Automation Mastery
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📘 CLASS NOTES

🧩 Module 2 · Core Nodes & Data Mastery

DAY 06 of 25

Code Node: JavaScript for Automators

"Jahan node khatam, wahan Code node shuru."

⚡ AAJ HUM YEH BANAYENGE

GST Invoice Calculator (Code Node)

Order ke items se subtotal, discount, GST (CGST+SGST) aur grand total JavaScript se calculate karke ek formatted invoice JSON file banata hai.

Code (JS) | Code (Python) | AI code helper

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Code node se student "no-code ki limit" tod deta hai. Thodi si JavaScript = unlimited power. AI se code likhwana bhi sikhaate hain.

🎯 Aaj ke baad aap yeh kar paoge

01 JavaScript essentials for n8n (no CS degree needed)

🌍 Recipe mein ingredients (variables), "agar namak kam ho to daalo" (if), "5 baar hilao" (loop).
⚙️ const total = $json.price * $json.qty; if (total > 1000) { ... }

02 Run once for all items vs each item

🌍 Poori class ki copies ek saath check vs har bachche ko alag bulana.
⚙️ All items: $input.all() · Each item: $json

03 Return format: [{ json: {...} }]

🌍 Courier ka fixed packing format — galat packing = reject.
⚙️ return [{ json: { total: 500 } }];

04 Array methods: map, filter, reduce

🌍 map = har aam ko kaatna, filter = kharab aam hatana, reduce = sabka juice ek glass mein.
⚙️ items.map(i => i.price), items.filter(i => i.qty > 0), items.reduce((s, i) => s + i.price, 0)

05 Using AI to write & debug Code node

🌍 Tuition teacher se homework samajhna — copy nahi, samajh ke likhna.
⚙️ Code node mein "Ask AI": "GST 18% add karo aur 2 decimal mein round karo".

06 Python in Code node — basics

🌍 Hindi aur English — dono se baat ho sakti hai.
⚙️ Python mode: for item in _input.all(): item.json["x"] = 1

⏱ Class ka plan (2 ghante)

10m	40m	40m	20m	10m

0:00-0:10 Recap

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

0:10-0:50 Concept: JS Crash Course

Variables, objects, loops — n8n context

🪄 Recipe likhna jaisa hai: pehle ingredients lo, phir step-by-step karo, end mein plate mein serve karo — yahi return hai.

0:50-1:30 Live Build

Invoice line-item calculator + GST

🪄 Aaj ki kahani yaad hai — 'Dukaan ka hisaab — calculator wala munshi'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

1:30-1:50 Quiz + Challenge

Module 2 quiz

🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

1:50-2:00 Q&A + Homework

🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Run once for all items

Saare items ek saath milte hain: $input.all()

🪄 **Analogy:** Poori class ki attendance ek baar mein

CONCEPT 02

Run once for each item

Har item par alag chalta hai: $json

🪄 **Analogy:** Har student se alag baat

CONCEPT 03

Return format

Hamesha array of {json: {...}} return karo.

🪄 **Analogy:** Packing format fixed hai

CONCEPT 04

map / filter / reduce

Arrays ke 3 superpowers — transform, chhaanto, jodo.

🪄 **Analogy:** Factory machines

🔑 Aaj ka main idea

Code node mein thoda JavaScript likh ke woh kaam karo jo 5–6 nodes mein hota. Bas 3 cheezein yaad rakho: data $input/$json se lo, map/filter/reduce se badlo, aur [{json: {...}}] return karo. AI se code likhwana bhi allowed hai — samajhna zaroori hai.

- map: har item ka price double karna
- filter: sirf 18+ age wale
- reduce: saare amounts ka total

📖 Ek kahani se samjho: Dukaan ka hisaab — calculator wala munshi

1. 🧾 **Order list aayi** = Webhook
2. 👩‍💻 **Munshi ne hisaab lagaya** = Code node
3. ➕ **Total + GST + discount** = reduce() / math
4. 🏷️ **Bill number diya** = Edit Fields
5. ↩️ **Customer ko bill** = Respond to Webhook

🪄 Ek line mein: “Recipe likhna jaisa hai: pehle ingredients lo, phir step-by-step karo, end mein plate mein serve karo — yahi return hai.”

🔬 Gehraai se samjho — har concept andar se

Kya hai, n8n andar kya karta hai, kab use karein, kab nahi, example aur common galti.

01 Run once for all items

📖 **Kya hai:** Code node ka yeh mode saare items ek saath, ek list mein deta hai. Code sirf ek baar chalta hai aur aap poori list par kuch bhi kar sakte ho — total, group, sort, duplicate hatana.

⚙️ **Andar kaise:** $input.all() = saare items ki list; har item ke andar .json mein data. $input.first() = pehla item. Aap jitne chahe utne items lauta sakte ho — 20 aaye aur 1 lautaya (total), ya 1 aaya aur 50 lautaye (split).

✅ **Kab use karein**

Jab ek item ka jawab doosre items par depend kare: total, average, rank, group by, duplicate hatana, sabse bada dhoondhna.

⛔ **Kab nahi**

Jab har item ka kaam alag aur simple ho (jaise sirf naam badalna) — tab Edit Fields ya "each item" mode zyada saaf hai.

🧪 **Example:** Calculate Invoice: $input.all().map(({ json: o }) => …) — har order ka poora invoice ek saath.

⚠️ **Common galti:** Is mode mein $json likhna — yahan $json kaam nahi karta (koi "current item" hota hi nahi). $input.all() ya $input.first().json use karo.

🪄 **All Items = poori class ek saath. $input.all() se list lo.**

02 Run once for each item

📖 **Kya hai:** Is mode mein Code node har item ke liye alag chalta hai. 10 items = 10 baar. Har baar $json mein bas wahi ek item hota hai.

⚙️ **Andar kaise:** Aap ek object lautate ho: return { json: { … } } — ya seedha return { …$json, naya: 1 }. n8n khud saare jawabon ko list bana deta hai. Paired item (kaunsa output kis input se bana) apne aap sahi rehta hai, isliye aage $("Node").item theek kaam karta hai.

✅ **Kab use karein**

Har item par same chhota calculation: GST, age from DOB, URL banana, text saaf karna.

⛔ **Kab nahi**

Total ya group jaisa kaam — kyunki ek baar mein sirf ek item dikhta hai, baaki nahi.

🧪 **Example:** return { json: { …$json, gst: +($json.price * 0.18).toFixed(2) } } → har product par GST juda.

⚠️ **Common galti:** "Each item" mode mein $input.all() se loop chala dena — 10 items × 10 = 100 items nikal aate hain.

🪄 **Each Item = har student se alag baat. $json se current item.**

03 Return format

📖 **Kya hai:** Code node se data lautane ka ek fixed format hai: items ki list, jahan har item { json: { ...fields } } hota hai. Files ho to { json, binary }.

⚙️ **Andar kaise:** n8n ko pata hona chahiye ki kitne items hain aur har item ka data kahan hai. Isliye [{ json: {...} }, { json: {...} }]. Nayi versions plain objects ki list ([{ a: 1 }]) bhi khud lapet deti hain, par { json } likhna sabse safe aur saaf hai. Kuch na lautana ho to return [].

✅ **Kab use karein**
Har Code node ke aakhir mein. Debug karte waqt output panel mein "N items" dekho — wahi aapka return hai.

⛔ **Kab nahi**
Date objects, functions ya undefined values lautane se bacho — JSON mein badal kar ajeeb dikhte hain. Date ho to .toISOString() ya Luxon toFormat().

🧪 **Example:** return [{ json: { invoice_no: "INV-1042", grand_total: 2886 } }];

⚠️ **Common galti:** return { total: 500 } (bina list) all-items mode mein — error: "Code doesn't return items properly". List [ ] mein lapeto.

🪄 **Hamesha [{ json: {...} }]. Packing ka format fixed hai.**

04 map / filter / reduce

📖 **Kya hai:** JavaScript lists (arrays) ke teen superpowers. map = har element ko badlo. filter = sirf kuch rakho. reduce = sabko jodkar ek value banao.

⚙️ **Andar kaise:** map(fn) utne hi elements lautata hai jitne mile (har ek badla hua). filter(fn) sirf woh rakhta hai jinke liye fn true de. reduce((total, x) => total + x, 0) shuru 0 se karke ek-ek jodta hai. Teeno original list ko nahi chhedte — nayi list dete hain.

✅ **Kab use karein**
Order ke line items ka total, sirf paid orders chunna, naamon ki comma wali list banana, items ko n8n format mein lapetna.

⛔ **Kab nahi**
Bahut lambe chained map().filter().map() — padhna mushkil. 2–3 steps tak theek; zyada ho to variables mein todo.

🧪 **Example:** const subtotal = o.items.reduce((s, i) => s + i.price * i.qty, 0); // 799×2 + 349×1 + 299×3 = 2844

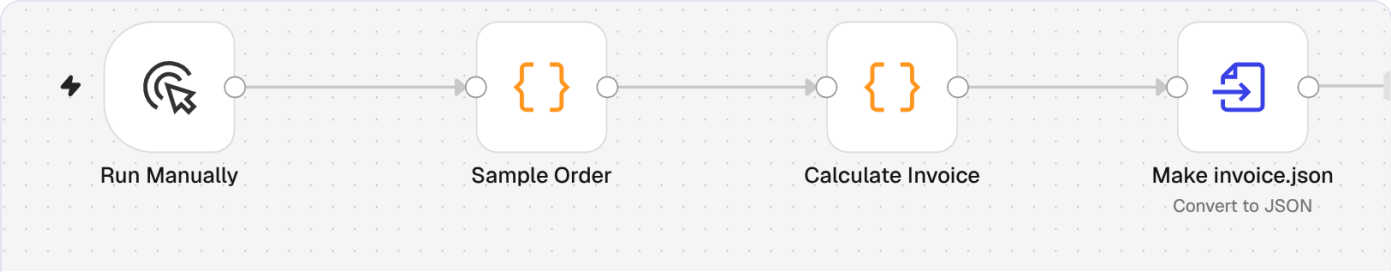
⚠️ **Common galti:** reduce mein shuru ki value (0) bhool jaana — khaali list par error aata hai aur objects ki list par ajeeb "[object Object]" jud jaata hai.

🪄 **map = badlo, filter = chhaanto, reduce = jodo (0 se shuru).**

🗺️ Aaj ka workflow: GST Invoice Calculator (Code Node)

Data left se right chalta hai. Har dabba ek n8n node hai.





📸 Asli n8n mein import karne par yeh dikhega. Laal ⚠️ = account (credential) connect karna baaki.

💰 Isse kitna kama sakte ho?

Shops, small manufacturers (GST invoice calculator) — aam taur par **₹5,000 – ₹15,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Webhook**
Order aata hai with line items

2 👨‍💻 **Code**
Har item ka subtotal, discount, 18% GST, grand total

3 ✏️ **Edit Fields (Set)**
invoiceNo = "INV-" + $now.toFormat("yyMMdd") + random

4 ⚡ **Respond to Webhook**
Calculated invoice JSON return

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Koi setup nahi — seedha Test karo!

✅ Test kaise karein

1. "Test workflow" dabao
2. "Calculate Invoice" output mein har line ka GST check karo
3. Last node mein "Download" se invoice.json lo

Data ka safar — har node ke baad

Aaj ke build mein data kaise badalta hai. Expressions samajhne ki chaabi yahi hai.

1 ⚡ Run Manually

Start signal — 1 khaali item.

```
{}
```

2 🔧 Sample Order

1 item = 1 order. Andar customer object aur 3 line items ki list.

```
{
  "order_id": "ORD-1042",
  "customer": {
    "name": "Sneha Kapoor",
    "state": "Delhi",
    "email": "sneha@example.com"
  },
  "coupon": "NEW10",
  "items": [
    {
      "sku": "TSHIRT-BLK",
      "name": "Black T-shirt",
      "qty": 2,
      "price": 799,
      "gst": 12
    },
    {
      "sku": "MUG-01",
      "qty": 1,
      "price": 349,
      "gst": 18
    },
    {
      "sku": "CAP-RED",
      "qty": 3,
      "price": 299,
      "gst": 12
    }
  ]
}
```

3 🔧 Calculate Invoice

NEW10 coupon = 10% off. Har line ka GST alag laga, fir sab joda. 2885.6 round hokar 2886.


```
{
  "invoice_no": "INV-1042",
  "date": "2026-12-03",
  "lines": [
    {
      "name": "Black T-shirt",
      "base": 1598,
      "discount": 159.8,
      "taxable": 1438.2,
      "cgst": 86.29,
      "sgst": 86.29,
      "total": 1610.78
    },
    "…(+2 lines)"
  ],
  "subtotal": 2844,
  "discount": 284.4,
  "cgst": 163,
  "sgst": 163,
  "grand_total": 2886
}
```


4 🔧 Make invoice.json

Data ab ek file ban gaya — Binary tab mein INV-1042.json. Wahan se Download karo ya aage Gmail/Drive ko do.


```
{
  "[binary] data": {
    "fileName": "INV-1042.json",
    "mimeType": "application/json",
    "fileExtension": "json"
  }
}
```

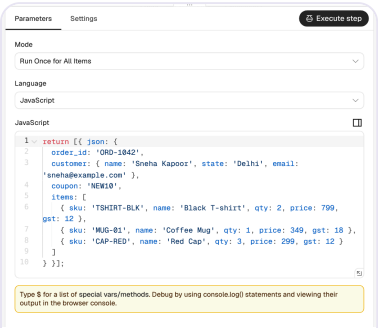

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Node anatomy — har setting ka matlab

Har zaroori option kya karta hai aur kab badalna hai. Isse aap khud naye workflows bana paoge.

Code — sample order

Ek nested order banata hai — customer object aur items ki list ke saath.

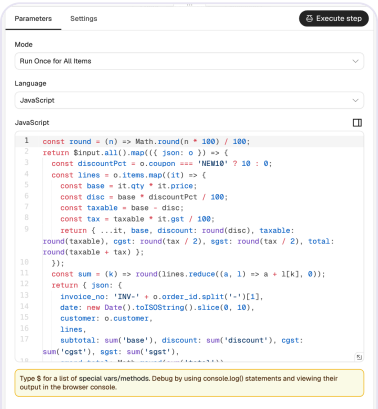


📸 Asli n8n: Code — sample order

Setting	Kya karti hai	Aaj ki value	Kab badlein
Mode	Run Once for All Items	`Run Once for All Items`	—
customer: { … }	Object ke andar object (nested)	`customer.name, customer.state`	Expression mein $json.customer.state
items: […]	Ek item ke andar line items ki list	`3 products`	Har product ka apna gst % (12 ya 18)
return [{ json: {…} }]	1 item = 1 order	`1 item`	Asli project mein Webhook / Shopify se aayega

Code — calculation

Aaj ka hero: discount, CGST/SGST aur grand total JavaScript se.

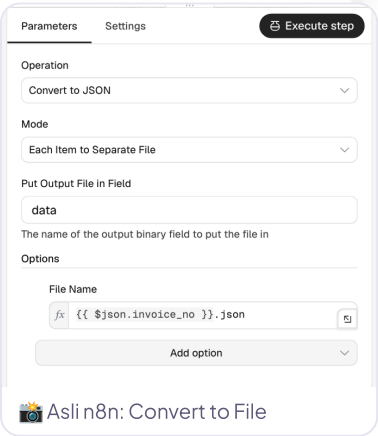


📸 Asli n8n: Code — calculation

Setting	Kya karti hai	Aaj ki value	Kab badlein
round = n => Math.round(n*100)/100	Paise tak (2 decimal) round	`86.292 → 86.29`	Paiso ka hisaab hamesha round karo
$input.all().map(({ json: o }) => …)	Har order par kaam; o = order ka data	`1 order`	Kai orders aaye to kai invoices
o.items.map(it => …)	Har line: base, discount, taxable, cgst, sgst, total	`3 lines`	GST ko 2 hisson mein: CGST + SGST (same state)
sum = k => lines.reduce(…)	Kisi bhi column ka total ek chhote function se	`sum("cgst") = 163`	Ek function, kai jagah use
grand_total: Math.round(…)	Final amount poore rupaye mein	`2885.6 → 2886`	Invoice par rounding line dikhana achha practice

Convert to File

JSON data ko ek downloadable file bana deta hai.



📸 Asli n8n: Convert to File

Setting	Kya karti hai	Aaj ki value	Kab badlein
Operation	Kis format mein: CSV, XLSX, JSON, HTML, Text, ICS, Base64…	`Convert to JSON`	Accountant ke liye XLSX
Mode	Each Item = har item ki alag file; All Items = sab ek file mein	`Each Item`	Report ke liye All Items
Options → File Name	File ka naam expression se	`{{ $json.invoice_no }}.json`	Naam mein date daalo: INV-1042-2026-12-03.json
Put Output File in Field	Binary field ka naam	`data (default)`	Gmail attachment / Drive upload mein yahi naam

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)

Calculator se har item jodna	10 min
GST alag nikaalna	5 min
Galti dobara check	5 min
	20 min

➜

⚡ Ab (n8n)

Code node 1 second mein

0 min

⏱ Har bill par 20 min bache, calculation galti zero.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
const o = $input.first().json;
const subtotal = o.items.reduce((s, i) => s + i.price * i.qty, 0);
const discount = subtotal * (o.discount / 100);
const gst = (subtotal - discount) * 0.18;
return [{ json: { subtotal, discount, gst: +gst.toFixed(2), total: +(subtotal - discount +
gst).toFixed(2) } }];
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "items": [
    {
      "name": "Course",
      "price": 9999,
      "qty": 1
    },
    {
      "name": "Templates",
      "price": 1999,
      "qty": 2
    }
  ],
  "discount": 10
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "subtotal": 13997,
  "discount": 1399.7,
  "gst": 2267.51,
  "total": 14864.81
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude

📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Order ke items se subtotal, discount, GST (CGST+SGST) aur grand total JavaScript se calculate karke ek formatted invoice JSON file banata hai.

Flow: Manual Trigger (Test run) → Code (Sample order) → Code (GST + discount + total) → Convert to File (invoice.json)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt

📋 Select karke copy karo

```
I'm building the n8n workflow "GST Invoice Calculator (Code Node)" (nodes: Run Manually, Sample Order, Calculate Invoice, Make invoice.json).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ Workflow JSON
4 nodes · ready to import
`adifyskills.com/workflows/day-06-gst-invoice-calculator.json`

🌐 Lesson page
Copy buttons, quiz aur video links
`adifyskills.com/day/6`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo

⚠️ Yeh galtiyan sabse common hain

- ! Object return karna array ki jagah
- ! "each item" mode mein $input.all() use karna
- ! Floating point rounding bhool jaana (toFixed)

⭐ Pro tips — time bachane wali tricks

- ★ Code node mein "Ask AI" se code generate karwao, phir samjho
- ★ console.log() output browser console mein dikhta hai
- ★ Complex logic ke liye pehle 5–6 nodes se banao, phir Code mein compress

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →



Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Order items se GST, discount aur total calculate karke formatted invoice JSON banana.

📝 Homework

3 tasks jo pehle 4–5 nodes mein hote the, unhe ek Code node mein likho.

- ☐ Homework ho gaya

BEGINNER

Naam ko uppercase karo Code se

☐

INTERMEDIATE

Duplicate emails Code se hatao

☐

ADVANCED

Indian number format (1,00,000) function likho

☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Agar subtotal ₹2,000 se upar ho to free shipping, warna ₹99 shipping jodo.

💡 Hint 1

Grand total ke calculation mein ek naya variable banao.

💡 Hint 2

const shipping = subtotal > 2000 ? 0 : 99; — yeh ternary operator hai.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🧪 Expression practice sheet

Input dekho aur khaali jagah mein expression likho. Jawab aakhri page par hain.

1. Order mein kitne products?

```
{"items":[{"name":"Black T-shirt"},{"name":"Coffee Mug"},{"name":"Red Cap"}]}
```

Mera jawab: _____

2. Saare product naam ek line mein (map)

```
{"items":[{"name":"Black T-shirt"},{"name":"Coffee Mug"},{"name":"Red Cap"}]}
```

Mera jawab: _____

3. Subtotal (reduce)

```
{"items":[{"price":799,"qty":2},{"price":349,"qty":1}]}
```

Mera jawab: _____

4. 12% GST wale kitne products? (filter)

```
{"items":[{"gst":12},{"gst":18},{"gst":12}]}
```

Mera jawab: _____

5. CGST (6%) 2 decimal tak

```
{"taxable":1438.2}
```

Mera jawab: _____

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Code node ka sahi return?

- (A) "a"
- (B) true
- (C) {a:1}
- (D) [{json:{a:1}}]

Q2. Saare items kaise milte hain?

- (A) $input.all()
- (B) $items
- (C) $all
- (D) $json

Q3. Sum nikaalne ke liye array method?

- (A) map
- (B) filter
- (C) reduce
- (D) sort

Q4. Code node kin languages mein?

- (A) PHP
- (B) Sirf JS
- (C) JS + Python
- (D) Sirf Python

📖 Aaj ke naye shabd

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

Webhook
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

JSON
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

Item
Data ki ek row/record. n8n mein data items ki list ki tarah flow karta hai.

Code Node
Jahan aap JavaScript likh ke apna custom logic chala sakte ho.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 7

HTTP Request & REST APIs Mastery

"Agar API hai, toh n8n se connect ho sakta hai."

☐ **Day 6 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → D
[{ json: { a: 1 } }]

Q2 → A
$input.all()

Q3 → C
reduce

Q4 → C
JS + Python

✅ Challenge ka solution

Calculate Invoice mein: const shipping = sum('base') > 2000 ? 0 : 99; aur grand_total mein + shipping, output mein shipping field bhi do.

🧪 Expression sheet ke jawab

1. `{{ $json.items.length }}` → **3**
2. `{{ $json.items.map(i => i.name).join(", ") }}` → **Black T-shirt, Coffee Mug, Red Cap**
3. `{{ $json.items.reduce((s, i) => s + i.price * i.qty, 0) }}` → **1947**
4. `{{ $json.items.filter(i => i.gst === 12).length }}` → **2**
5. `{{ ($json.taxable * 0.06).toFixed(2) }}` → **86.29**

📚 Official docs

- Code node ↗
- Cookbook ↗
- MDN JS Guide ↗
- Cookbook: Code node ↗
- AI coding in Code node ↗
- MDN: Array.reduce ↗

▶ Videos

- n8n code node javascript tutorial ↗
- n8n code node python ↗

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